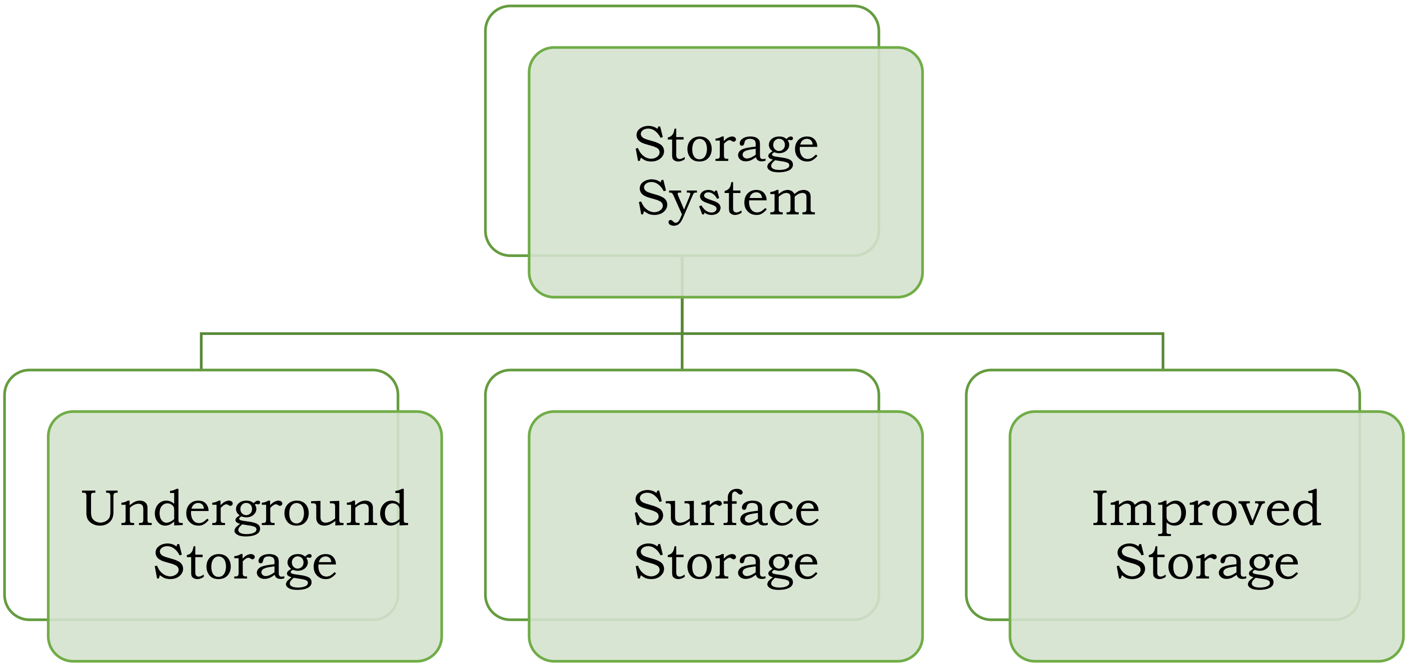
FOOD STORAGE SYSTEM

Storage

Storage is an important marketing function, which invo**lves holding and preserving goods from the time they are produced** until they are needed for consumption.

- The storage of goods, therefore, from the time of production to the time of consumption, ensures a **continuous flow of goods in the market**.
- **Storage protects the quality of perishable and semi-perishable products** from deterioration;
- Some of the goods e.g., woollen garments, **have a seasonal demand**. To cope with this demand, production on a continuous basis and storage become necessary;
- It helps in the **stabilization of prices by adjusting demand and supply**;
- Storage **is necessary for some period for performance** of other marketing functions.
- Storage **provides employment and income** through price advantages.

Types Of Storage



1. Underground Storage Structures

Underground storage **structures are dugout structures similar to a well** with sides **plastered with cow dung**. They may also be lined with stones or sand and cement. They may be circular or rectangular in shape. The capacity **varies with the size of the structure**.

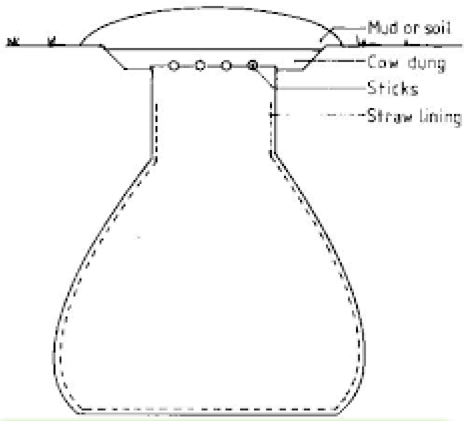


Figure 1- Underground Storage System

Advantages

- Underground storage structures **are safer** from threats from various external sources of damage, such as theft, rain or wind.
- The **underground storage space can temporarily be utilized** for some other purposes with minor adjustments; and
- The underground storage structures **are easier to fill up** owing to the factor of gravity.

2. Surface storage structures

Foodgrains in a ground surface structure can be stored in two ways

- **Bag Storage**
- **Bulk Storage**

A. Bag Storage

Each bag **contains a definite quantity**, which can be bought, sold or dispatched without difficulty;

Figure 2- Bag Storage

- Bags are **easier to load or unload**.
- It is easier to keep separate **lots with identification marks** on the bags.
- The bags which are identified as **infested on inspection can be removed** and treated easily; and

- The problem of the sweating of grains does not arise because the surface of the bag is exposed to the atmospheres.

B. Bulk or loose storage

Advantages

- The **exposed peripheral surface area** per unit weight of grain is less. Consequently, the danger of damage from external sources is reduced.
- Pest infestation **is less because of almost airtight conditions** in the deeper layers.
- The government of **India has made efforts to promote improved storage** facilities at the farm level.

Figure 3- Loose Storage

C. Improved Grain Storage Structures

Improved grain storage is of two type –

1. **For small scale storage**
2. **For large scale storage**

1. For Small Scale Storage

PAU bin- This is a **galvanized metal iron structure**. It's capacity ranges from 1.5 to 15 quintals.

Pusa bin- This is a **storage structure is made of mud or bricks with** a polythene film embedded within the walls.

<u>Hapur Tekka</u>- It is a **cylindrical rubberised cloth structure supported by bamboo poles** on a metal tube base, and has a small hole in the bottom through which grain can be removed.

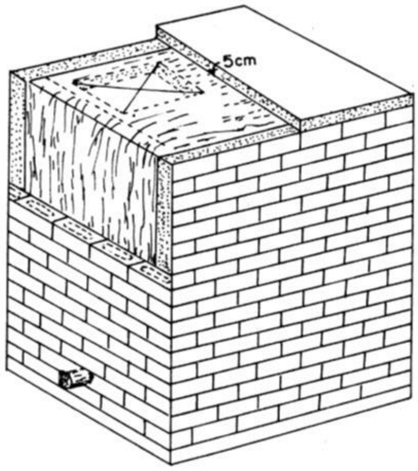


Figure 4 – PUSA Bin

2. <u>For Large Scale Storage</u>

Figure 5 – Cover And Plinth Storage

<u>CAP Storage (Cover and Plinth)-</u> It involves the construction of **brick pillars to a height of 14" from the ground**, with grooves into which **wooden crates are fixed for the stacking of bags of foodgrains**. The structure can be fabricated in less than 3 weeks. It is an economical way of storage on a large scale.

<u>Silos</u>- In these structures, the **grains in bulk are unloaded on the conveyor belts and**, through mechanical operations, are carried to the storage structure. The storage **capacity of each of these silos is around 25,000 tonnes**.

Figure 6 – Silo Storage of Grains

Warehousing

Warehouses are scientific storage structures especially constructed for the protection of the quantity and quality of stored products.

Figure 7-Warehouse

Importance

1. **Scientific storage**-The product is **protected against quantitative and qualitative losses** by the use of such methods of preservation as are necessary.

2. **Financing-**Warehouses meet the **financial needs of the person who stores the product**. Nationalized banks advance credit on the security of the warehouse receipt issued for the stored products to the extent of 75 to 80% of their value.

3. **Price Stabilization**-**Warehouses help in price stabilization** of agricultural commodities by checking the tendency to making post-harvest sales among the farmers.
4. **Market Intelligence**-**Warehouses also offer the facility of market information** to persons who hold their produce in them.

Working of Warehouses

Acts: - The warehouses (**CWC and SWCs**) work under the **respective Warehousing Acts** passed by the Central or State Govt

Eligibility: - Any person **may store notified commodities** in a warehouse on agreeing to pay the specified charges.

Warehouse Receipt (Warrant): - This is receipt/warrant issued by the warehouse manager/owner to the person storing his produce with them. **This receipt mentions the name and location of the warehouse, the date of issue, a description of the commodities**, including the grade, weight and approximate value of the produce based on the present prices.

Use of Chemicals: - The produce accepted at the warehouse is **preserved scientifically and protected against rodents, insects and pests and other infestations**. Periodical dusting and fumigation are done at the cost of the warehouse in order to preserve the goods.

Financing - The warehouse **receipt serves as a collateral security** for the purpose of getting credit.

Delivery of produce: - The warehouse receipt has to be **surrendered to the warehouse owner before the withdrawal of the goods**. The holder may take delivery of a part of the total produce stored after paying the storage charges.

Types Of Warehouses

1. On the basis of Ownership

Private warehouses: These are **owned by individuals, large business houses** or wholesalers for the storage of their own stocks. They also store the products of others.

Public warehouses: These are the warehouses, which are **owned by the govt.** and are meant for the storage of goods.

Bonded warehouses: These warehouses **are specially constructed at a seaport or an airport and accept imported goods for storage till the payment of customs** by the importer of goods. These **warehouses are licensed by the govt. for this purpose**. The goods stored in this warehouse are bonded goods.

Following services are rendered by bonded warehouses

- The importer of goods is saved from the **botheration of paying customs duty** all at one time because he can take delivery of the goods in parts.
- The operation necessary for the **maintenance of the quality of goods** - spraying and dusting, are done regularly.
- **Entrepot trade (re-export of imported goods**) becomes possible.

2. On the basis of Type of Commodities Stored

General Warehouses: These are **ordinary warehouses** used for storage of most of foodgrains, fertilizers, etc.

Special Commodity Warehouses: These are warehouses, which are specially constructed for the storage of **specific commodities like cotton, tobacco, wool and petroleum products**.

Refrigerated Warehouses: These are warehouses in which temperature is maintained as per requirements and are meant for **such perishable commodities as vegetables, fruits, fish, eggs and meat.**

Warehousing in India

1. Central warehousing corporation (CWC)

This corporation was **established as a statutory body in New Delhi on 2nd March 1957**. The Central Warehousing Corporation provides safe and reliable storage facilities for about **120 agricultural and industrial commodities.**

Functions

- To acquire and **build godowns and warehouses** at suitable places in India.

- To run warehouses for the **storage of agricultural produce**, **seeds, fertilizers** and **notified commodities** for individuals, co-operatives and other institutions,
- To act as an **agent of the govt. for the purchase, sale, storage** and distribution of the above commodities.
- To arrange **facilities for the transport** of above commodities.
- To subscribe to the **share capital of state Warehousing** corporations and
- To carry **out such other functions** as may be prescribed under the Act.

The Central Warehousing Corporation is running air-conditioned godowns at Calcutta, Bombay and Delhi, and provides cold storage facilities at Hyderabad.

Special storage facilities have been provided by **the Central Warehousing Corporation for the preservation of hygroscopic and fragile commodities**.

The corporation has also **evolved techniques for the storage of spices, coffee, seeds and other commodities.**

2. State Warehousing Corporations (SWCs)

Separate warehousing corporations were also set up in different States of the Indian Union.

- The areas of operation of the State Warehousing Corporations **are centres of district importance**.
- **The total share capital** of the State Warehousing Corporations is **contributed equally by the concerned State Govt. and the Central Warehousing Corporation.**

3. Food corporation of India (FCI)

he Food Corporation of India **has also created storage facilities**. The Food Corporation of **India is the single largest agency for storage of agro produce**

The Food Corporation of India **was setup under the Food Corporations Act 1964, in order to fulfil following objectives of the Food policy**

- **Effective price support operations** for safeguarding the interests of the farmers.

- **Distribution of food grains throughout the country** for Public Distribution System; and
- **Maintaining satisfactory level of operational and buffer stocks of food grains** to ensure National Food Security.

4. Warehousing Development and Regulatory Authority (WDRA)

WDRA constituted **on 26.10.2010** under the **Warehousing (Development and Regulation) Act, 2007 vide Government of India Gazette Notification dated 26th October 2010**.

The **WDRA is a Statutory Authority under the Department of Food and Public Distribution, Government of India**.

The Authority has its **Headquarters in New Delhi.**

The Act provides for the establishment of the WDRA to exercise the powers conferred on it and to perform the functions assigned to it under the Act, Rules and **Regulations for the development and regulation of warehouses, negotiability of warehouse receipts and promote orderly growth of the warehousing business in the country.**

A New initiative under WDRA for promotion of post-harvest storage and warehousing

1. e-Kisan Upaj Nidhi

- **e-Kisan Upaj Nidhi provides the farmers, an online platform** to obtain **post-harvest loans from banks** by pledging their electronic Negotiable Warehouse Receipts (e-NWRs) for the stocks kept in WDRA registered warehouses.
- A **joint endeavour of Department of Food and Public Distribution, WDRA, Department of Financial Services and NABARD**,
- This gateway will **improve ease of obtaining pledge finance** by farmers against their stocks.

Cold Storage

India is the largest producer of fruits and second largest producer of vegetables in the world. In spite of that per capita availability of fruits and vegetables **is quite low because of post-harvest losses which account for about 25% to 30% of production**.

- Besides, quality of a sizable quantity of produce also **deteriorates by the time it reaches the consumer**.
- Most of the problems relating to the marketing **of fruits and vegetables can be traced to their perishability.**
- **Perishability is responsible for high marketing costs**, market gluts, price fluctuations and other similar problems.

with a view to ensuring the observance of proper conditions in the cold stores and to providing for development of the industry in **a scientific manner**, **the govt of India and the ministry of agriculture promulgated an order known as "Cold Storage Order, 1964" under Section 3 of the Essential Commodities Act,1955**.

- **The Agricultural Marketing Advisor to the Govt of India is the Licensing Officer**.
- A cold storage **facility accessible to them will go a long way in removing the risk of distress** sale to ensure better returns.

Status Of Cold Storage And Its Potential In India

Due to diverse agro climatic conditions and better availability of package of practices, the production is gradually rising. Although, there is a vast scope for increasing the production, the **lack of cold storage and cold chain facilities are becoming major bottlenecks in tapping the potential**.

The cold storage facilities now available are mostly **for a single commodity like potato, orange, apple, grapes, pomegranates, flowers, etc. which results in poor capacity utilization**.

Storage of foods and Storage Conditions

Foods and many other commodities can be preserved by storage **at low temperature, which retards the activities of microorganisms**. Microorganisms are the spoilage agents and **consist of bacteria, yeasts and molds.**

- Low temperature **does not destroy those spoilage agents as does high temperature, but greatly reduces their activities**.

- The low temperature necessary for preservation depends on the storage time required often referred **to as short- or long-term shortage and the type of product**.

Figure 8 – Cold Storage

In general, there are three groups of products:

1. **Foods that are alive at the time of storage**, distribution and sale **e.g. fruits and vegetables,**
2. **Foods that are no longer alive and have been processed** in some form e**.g. meat and fish products**
3. **Commodities that benefit from storage at controlled temperature** e**.g. beer, tobacco, khandsari, etc.**

The Best Method of Preserving These Items

- Best method of preserving live product is to **keep the product alive and** at the same time **retard the natural enzyme activity which will retard the rate of ripening or maturity**.
- Preservation of non-living foods is more difficult since they are susceptible to spoilage. **Long term storage of meat and fish product can only be achieved by freezing** and **then by storing it at temperature below -15°C.**

- Only certain fruits and vegetables can benefit from freezing. However, **for fruits and vegetables one should be very careful about the recommended storage temperature and humidity a deviation** from which will have adverse effect on the stored product leading to even loss of the entire commodity.
- Products such as apples, tomatoes, oranges, etc. cannot be frozen and close control of temperature is necessary for long term storage.
- Generally, fruit **and vegetable product is stored under controlled atmosphere and modified atmosphere conditions**.
- Dairy products are produced **from animal fats and therefore non-living foodstuffs**. They suffer from the oxidation and breakdown of their **fats, causing rancidity**. Packaging to **exclude air and hence Oxygen can extend storage life** of such foodstuffs.

Technology

A cold storage unit incorporates a refrigeration system to maintain the desired room environment for the commodities to be stored.

A Refrigeration System Works on Two Principles:

- **Vapour Absorption System (VAS)-** VAS, although comparatively costlier, is quite economical in operation and adequately compensates the higher initial investment. In the **vapor absorption system, the refrigerant used is ammonia with water** as the absorber, or, **water if used as a refrigerant with lithium bromide as the absorber**. This cycle does not have a compressor.

- **Vapour Compression System (VCS)**- VCR stands for Vapour Compression Refrigeration Cycle. It is **the most widely used refrigeration system in which the working fluid is a vapor which readily evaporates and condenses** (changes alternatively between the vapor and liquid phase) without leaving the refrigerating plant. In a refrigeration system,

Function of refrigerants

Refrigerants are used to pick up heat by evaporation at a **lower temperature and pressure from the storage space and give up the heat by condensation at a higher temperature and pressure in a condenser**.

Freon used to be a common refrigerant but as it causes environmental degradation, its use is going to be banned by the year 2008. **Therefore, Ammonia is being increasingly used** and preferred for horticultural and plantation produce in cold storage units.

Table – Required Temperature And Relative Humidity For Different Fruit And Vegetables

Commodity	Temperature (°C)	Relative Humidity (%)
Potato	1.5 - 4	90 - 94
Grapes	-1 - 1	85 - 90
Apple	-1 - 3	90 - 98
Mango	11 - 18	85 - 90
Cabbage	0 - 2	90 - 95

Last year Mains question from this section

Essay in English descriptive

Discuss the bottlenecks and solution in supply chain management of fruits and vegetables in India